

classify_to_functional_type.R

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Classify Community Data to Functional Types

Description

Classifies taxa in a long-format community data frame to functional-type (FT) labels using a pre-computed FT classification table. Aggregates pollen proportions by ‘(dataset_name, age, functional_type)’. Produces the same output column structure as ‘classify_taxonomic_resolution()’ so it can be used as a drop-in replacement downstream.

Usage

```
classify_to_functional_type(data, data_ft_classification, verbose = TRUE)
```

Arguments

data	A data frame containing community data with columns ‘taxon’, ‘dataset_name’, ‘age’, and ‘pollen_prop’.
data_ft_classification	A data frame mapping taxa to functional types. Must contain columns ‘taxon_name’ (character) and ‘functional_type’ (integer). Typically produced by ‘cluster_functional_types()’ and loaded via ‘get_functional_type_classification()’.
verbose	Logical. If ‘TRUE’ (default), informational progress messages are printed to the console via ‘cli’. Note: the warning issued when taxa are absent from ‘data_ft_classification’ is always emitted regardless of this flag.

Details

Steps performed:

1. Validate arguments.
2. Left-join 'data' to 'data_ft_classification' on 'data\$taxon == data_ft_classification\$taxon_name'.
3. Drop unmatched taxa (NA functional type) with a warning.
4. Create 'taxon' labels "FT_functional_type".
5. Aggregate 'pollen_prop' by '(dataset_name, age, taxon)'.
6. Full-join back to a '(dataset_name, age, taxon)' cross-reference to preserve true negatives.

Value

A data frame with the same column names as 'data'. The 'taxon' column is replaced by functional-type labels of the form "FT_1", "FT_2", etc. 'pollen_prop' is aggregated (summed) by '(dataset_name, age, taxon)'. All '(dataset_name, age)' combinations present in 'data' are preserved (true negatives kept via a cross-reference join). Taxa not found in 'data_ft_classification' are dropped with a 'cli::cli_warn()' message.

See Also

[classify_taxonomic_resolution()], [get_functional_type_classification()], [cluster_functional_types()]

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